

Template bank for subsolar mass compact binary mergers in the fourth observing run of Advanced LIGO, Advanced Virgo, and KAGRA

Chad Hanna,^{1,2,3,4,*} James Kennington^{1,2,*} Wanting Niu^{1,2,*} Shio Sakon^{1,2,*} Divya Singh,^{1,2,5,*} Shomik Adhichary^{1,2}
 Pratyusava Baral,⁶ Amanda Baylor⁶ Kipp Cannon⁷ Sarah Caudill,^{8,9} Bryce Cousins,^{10,1,2} Jolien D. E. Creighton^{1,2}
 Becca Ewing,^{1,2} Heather Fong,^{11,7,12} Richard N. George¹³ Patrick Godwin^{14,1,2} Reiko Harada^{7,12} Yun-Jing Huang,^{1,2}
 Rachael Huxford,^{1,2} Prathamesh Joshi^{1,2} Soichiro Kuwahara^{7,12} Alvin K. Y. Li^{7,12} Ryan Magee¹⁵
 Duncan Meacher⁶ Cody Messick,⁶ Soichiro Morisaki¹⁶ Debnandini Mukherjee^{17,18} Alexander Pace,^{1,2}
 Cort Posnansky^{1,2} Anarya Ray,^{6,19} Surabhi Sachdev,^{20,6} Stefano Schmidt^{21,22} Urja Shah²⁰ Ron Tapia,^{1,4}
 Leo Tsukada^{1,2,23,24} Koh Ueno,⁷ Aaron Viets²⁵ Leslie Wade,²⁶ Madeline Wade,²⁶
 Zach Yarbrough,²⁷ and Noah Zhang²⁰

¹*Department of Physics, The Pennsylvania State University, University Park, Pennsylvania 16802, USA*

²*Institute for Gravitation and the Cosmos, The Pennsylvania State University, University Park, Pennsylvania 16802, USA*

³*Department of Astronomy and Astrophysics, The Pennsylvania State University, University Park, Pennsylvania 16802, USA*

⁴*Institute for Computational and Data Sciences, The Pennsylvania State University, University Park, Pennsylvania 16802, USA*

⁵*Department of Physics, University of California, Berkeley, California 94720, USA*

⁶*Leonard E. Parker Center for Gravitation, Cosmology, and Astrophysics, University of Wisconsin-Milwaukee, Milwaukee, Wisconsin 53201, USA*

⁷*RESCEU, The University of Tokyo, Tokyo, 113-0033, Japan*

⁸*Department of Physics, University of Massachusetts, Dartmouth, Massachusetts 02747, USA*

⁹*Center for Scientific Computing and Data Science Research, University of Massachusetts, Dartmouth, Massachusetts 02747, USA*

¹⁰*Department of Physics, University of Illinois, Urbana, Illinois 61801 USA*

¹¹*Department of Physics and Astronomy, University of British Columbia, Vancouver, British Columbia V6T 1Z4, Canada*

¹²*Graduate School of Science, The University of Tokyo, Tokyo 113-0033, Japan*

¹³*Center for Gravitational Physics, University of Texas at Austin, Austin, Texas 78712, USA*

¹⁴*LIGO Laboratory, California Institute of Technology, MS 100-36, Pasadena, California 91125, USA*

¹⁵*LIGO Laboratory, California Institute of Technology, Pasadena, California 91125, USA*

¹⁶*Institute for Cosmic Ray Research, The University of Tokyo, 5-1-5 Kashiwanoha, Kashiwa, Chiba 277-8582, Japan*

¹⁷*NASA Marshall Space Flight Center, Huntsville, Alabama 35811, USA*

¹⁸*Center for Space Plasma and Aeronomic Research, University of Alabama in Huntsville, Huntsville, Alabama 35899, USA*

¹⁹*Center of Interdisciplinary Education and Research in Astrophysics, Northwestern University, Illinois 60201, USA*

²⁰*School of Physics, Georgia Institute of Technology, Atlanta, Georgia 30332, USA*

²¹*Nikhef, Science Park 105, 1098 XG, Amsterdam, The Netherlands*

²²*Institute for Gravitational and Subatomic Physics (GRASP), Utrecht University, Princetonplein 1, 3584 CC Utrecht, The Netherlands*

²³*Department of Physics and Astronomy, University of Nevada, Las Vegas, 4505 South Maryland Parkway, Las Vegas, Nevada 89154, USA*

²⁴*Nevada Center for Astrophysics, University of Nevada, Las Vegas, Nevada 89154, USA*

²⁵*Concordia University Wisconsin, Mequon, Wisconsin 53097, USA*

²⁶*Department of Physics, Hayes Hall, Kenyon College, Gambier, Ohio 43022, USA*

²⁷*Department of Physics and Astronomy, Louisiana State University, Baton Rouge, Louisiana 70803, USA*



(Received 18 December 2024; accepted 30 May 2025; published 8 August 2025)

Matched-filtering searches for gravitational-wave signals from compact binary mergers employ template banks which are a collection of modeled waveforms described by unique intrinsic parameters. We present

*These authors contributed equally to this work.

two banks designed for low-latency and archive subsolar mass (SSM) searches in data from the fourth observing run of LIGO-Virgo-KAGRA, and demonstrate the efficacy of the banks via simulated signals. Further, we introduce several new methodological developments to the geometric placement algorithm [C. Hanna *et al.*, *Phys. Rev. D* **108**, 042003 (2023); C. Hanna, MANIFOLD (2024)], including a more stable metric-estimation technique and a boundary-padding procedure. Together, these enhancements enable the method to work for exceedingly low component masses necessary for SSM bank production, representing the first successful application of the MANIFOLD-method to lower-mass regions of the compact binary parameter space. The archive search bank contains a total of 3,452,006 templates, and covers a mass parameter space of 0.2 to $10M_{\odot}$ in the larger component and 0.2 to $1.0M_{\odot}$ in the smaller component, the spin parameter space of -0.9 to 0.9 for masses above $0.5M_{\odot}$ and -0.1 to 0.1 for masses below $0.5M_{\odot}$, and the mass ratio parameter space of 1 to 10 . The power spectral density used was from a week of the first half of the fourth observing run of Advanced LIGO, Advanced Virgo, and KAGRA, and the low frequency cutoff was set to 45 Hz with a maximum waveform duration of 128 secs. The bank simulations performed using SBank have shown that the banks presented in this paper have sufficient efficacy for use in their respective searches.

DOI: [10.1103/c97v-bmj8](https://doi.org/10.1103/c97v-bmj8)

I. INTRODUCTION

The first gravitational wave (GW) detection of a binary black hole (BBH) merger, GW150914 [1–3], by Advanced Laser Interferometer Gravitational-Wave Observatory (LIGO) [4–8] and Advanced Virgo (aVirgo) [9,10], brought about the era of GW astronomy which has been furthered by subsequent detections of more BBH mergers, binary neutron star (BNS) mergers [11,12] and neutron star–black hole binary (NSBH) mergers [13]. These observations have provided insight into the mass spectrum and population properties for merging compact objects along with the equation of state of dense matter [14,15].

Since the observing run (O1), the LIGO Scientific, Virgo, and KAGRA Collaborations (LVK) [16–18] has published over 90 GW events from modeled and unmodeled searches for compact binary coalescences (CBCs) [19]. The GstLAL-based compact binary search pipeline [20–27] is one of the matched-filtering [28–31] pipelines [32–39] which runs low-latency searches for CBCs by correlating the data with banks of templates that model the GW signals expected from these binaries. Beyond the searches for CBCs with component masses greater than a solar mass, searches for compact binaries with at least one component below $1M_{\odot}$ have also been carried out in previous observing runs using data from Initial LIGO [40,41], and Advanced LIGO and aVirgo [42–50], henceforth LIGO and Virgo respectively, as well as in the ongoing observing run.

Since stellar processes are expected to produce only super-solar mass compact objects, searches for GWs from mergers of subsolar-mass (SSM) components are motivated by formation channels that extend beyond the standard model, including dark matter models that predict the existence of compact objects with masses below $1M_{\odot}$ and primordial black holes (PBHs) that formed from overdensities in the early universe. Although the mass spectrum of PBHs can coincide with the spectrum of stellar BHs, the existence of BHs below $1M_{\odot}$ is not excluded [51,52].

The first BBH event detected by LIGO was hypothesized to be of primordial origin probing the long-standing question about the nature of dark matter [53–55]. Other scenarios like the cooling and gravitational collapse of dissipative dark matter halos are predicted to produce SSM BHs with appreciable detectable merger rates [56,57]. GW observations or lack thereof, especially of SSM mergers, can constrain PBH and dispasitive black hole binaries (DBH) populations and their dark matter abundances [44,45,52,56,58]. Other exotic objects like boson stars [59] and formation channels like imploding neutron stars [60–65] from the dark matter sector can populate the BH mass spectrum around a solar mass and also fall within the parameter space of SSM searches. While there have been no detections of GWs from SSM systems, the most recent SSM search by the LVK using data from second half of the third observing run (O3b) [45] have provided upper limits on the merger rate of compact binaries with at least one SSM component.

LVK’s matched-filtering pipelines have operated super-solar and SSM searches using template banks produced by stochastic template placing algorithms in previous observing runs [19,42–45,66,67]. In fourth observing run (O4), individual LVK matched-filtering pipelines produced independent template banks using power spectral densities (PSDs) that better reflect O4 detector sensitivities. GstLAL used a binary-tree approach template placement algorithm [68,69], MANIFOLD, MANIFOLD to produce its template banks for O4 [26], including the SSM template banks, to improve the computational efficiency of producing the template bank and downstream data products derived from the template bank. Producing SSM template banks with MANIFOLD required development work described in this paper, as the low-mass region was not investigated with MANIFOLD previously. In this paper, we motivate using MANIFOLD for the O4 SSM template bank generation in Sec. II, present the methods to produce template banks that

extend its component masses down to $0.2M_{\odot}$ in Sec. IV and Appendix A, describe the parameter space covered by the O4 GstLAL low-latency SSM search in Sec. III, illustrate the template bank effectualness test results in Sec. V, and provide the template bank parameter descriptions and bank effectualness test results of the archival SSM search in the Appendix B. Through the template bank effectualness tests, we show that the SSM template banks are capable of detecting potentially electromagnetically bright systems with low masses, which provides opportunities for multi-messenger detections when SSM searches are deployed.

II. MOTIVATION

In this section, we motivate the generation of new template banks with a computationally efficient method of template bank generation, MANIFOLD [68], for GstLAL's searches in O4 by addressing the following; the use of noise curves that better reflects O4 detector sensitivity, the computational efficiency of generating template banks, the computational efficiency of producing data products derived from the template banks, and the need for low-latency SSM template banks. Here, SSM binaries consist of at least one SSM component.

First, GstLAL template banks for O4 were newly generated instead of reusing template banks from previous observing runs, as the sensitivities of the detectors have changed compared to third observing run (O3) and because the sensitivities of the detectors dictate the discreteness of the template placement. Unlike previous observing runs where a common template bank was used across pipelines for each observing run [19,42–45,66,67], pipelines produced independent template banks in O4. GstLAL's supersolar mass template bank [26] and the low-latency SSM template bank (see Appendix B) used the projected O4 PSD, as they were generated prior to the start of O4. The SSM archival search (i.e., “offline” search), template bank for GstLAL was generated using a portion of the first half of the fourth observing run (O4a) PSD as it was produced after obtaining O4a data.

Second, MANIFOLD was used to generate GstLAL's O4 template banks due to its computational efficiency in producing templates. As shown in Sakon *et al.* [26], MANIFOLD reduces the computational cost of generating template bank significantly compared to traditional stochastic template placement methods, e.g., from $\mathcal{O}$ (week) to $\mathcal{O}$ (10 minutes) for supersolar mass template banks. Such abilities to rapidly generate template banks enable developers to test and tune template bank configurations in shorter timescales than using stochastic methods.

Third, the generation of downstream data products that are derived from the template bank, such as the population model [70] and p_{astro} model [27], are computationally efficient with MANIFOLD as the template volume elements needed to compute the population model and p_{astro} model are stored in the process of generating the template banks.

The binary tree algorithm of MANIFOLD enables computationally efficient checkerboarding of the template bank, which is a procedure that splits the template bank into two nearly identical halves such that two instances of analyses can be run on separate high-throughput computing clusters to mitigate failures [26].

Fourth, a computationally efficient SSM template bank for low-latency SSM searches is needed in O4 since for the first time in the history of GW detections, the LVK is searching for SSM systems in low-latency. Due to the number of available computational resources that GstLAL has for low-latency analyses, GstLAL used a limited-parameter space template bank that extends down to $0.5M_{\odot}$ and spins less than 0.3 in magnitude for its low-latency analysis whereas its archival search template bank extended down to component masses of $0.2M_{\odot}$ and spins spanned up to 0.9 in magnitude, which matches SSM template banks of previous observing runs. Template banks extending down to low-mass regions have not been generated with MANIFOLD before and the original version of MANIFOLD described in Hanna *et al.* [68] required new methods to produce a SSM template bank with minimum component masses of $0.2M_{\odot}$ while limiting the number of templates to $\mathcal{O}(10^6)$ and overcoming numerical and computational issues. Such new methods, namely the metric estimation stability improvement and the boundary enhancement are illustrated in Sec. IV. Due to these reasons, all template banks used by GstLAL in O4 have been generated with MANIFOLD, in addition to having consistency and compatibility across the O4 template banks.

III. DESIGN

The offline SSM template bank covers the same parameter space as the SSM template bank used in the O3 [44,45]. This ensures consistency with previous searches where the target search space was determined with consideration for the computational cost of SSM searches. Refer to Appendix B for the design and bank validity test results of the low-latency SSM template bank.

Table I lists the parameter and configuration choices that were used to generate the template bank. Following previous LVK SSM searches, we define SSM binaries as those that contain at least one component whose mass is below $1M_{\odot}$. The templates are limited to spin-aligned systems to reduce computational costs and preserve consistency with previous searches.¹ The z -components contribute to the effective spin, χ_{eff}

¹There have been studies on the impact of precession searches on the search sensitivities using the GstLAL algorithm and a modified version of MANIFOLD [71,72]. These studies showed an exponential increase in the number of templates. The SSM searches presented in this paper exclude precession due to limited computational resources and to maintain consistency with the previous LVK SSM searches.

TABLE I. Parameter space of the O4 offline SSM template bank.

Parameter	Offline SSM
Primary mass, m_1	$\in [0.2, 10M_\odot]$
Secondary Mass, m_2	$\in [0.2, 1M_\odot]$
Mass ratio, $q = m_1/m_2$	$\in [1, 10]$
Dimensionless spin, $s_{1,z}$, for $m_i \in [0.2, 0.5M_\odot]$	$ s_{i,z} < 0.1$
Dimensionless spin, $s_{1,z}$, for $m_i > 0.5M_\odot$	$ s_{i,z} < 0.9$
Lower frequency cutoff	45 Hz
Higher frequency cutoff	1024 Hz
Waveform approximant	IMRPhenomD
Minimum match	96.5%
Maximum waveform duration	128 secs
Total number of templates	3,452,006

$$\chi_{\text{eff}} = \frac{m_1 \times s_{1,z} + m_2 \times s_{2,z}}{m_1 + m_2}. \quad (1)$$

The MANIFOLD method [73] assigns the same z -component spins for both components in the binary such $s_{1,z} = s_{2,z}$, and the value of χ_{eff} is treated as $s_{i,z}$. The restricted component spins for low mass binaries was chosen to primarily reduce computational costs of filtering data by restricting the size of the bank [44,45]. Since low-mass templates have long waveforms that are inspiral-dominated, we restrict the lower frequency cutoff to 45 Hz for computational efficiency and to be consistent with previous LVK SSM searches [42–45]. We further set a maximum duration cutoff [26] at 128 secs, which is a feature to set a limit on the template duration by truncating the low-frequency end for to reduce memory usage. Depending on the intrinsic parameters, the lower frequency cutoff on some templates in the bank can be higher than 45 Hz. We apply this feature both in the process of computing matches for template placement [68] and in the process of performing SVD waveform [26], whose impact on SNR recovery is discussed in Sec. V. 128 seconds is chosen as a number of the power of two closest to the duration of BNS waveforms that are detectable in the LIGO sensitivity band [11]. We estimated the signal-to-noise ratio (SNR) loss due to the maximum duration cutoff by comparing the optimal SNRs obtained with and without the maximum duration requirement. The optimal SNR without the maximum duration cutoff was computed using $f_{\text{low}} = 45$ Hz. The SNR loss increasingly affects systems with $\mathcal{M} \lesssim 0.4M_\odot$, reaching a loss of nearly 16% for systems with $m_i = 0.2M_\odot$ ($i = 1, 2$). Templates with $m_i \gtrsim 0.5M_\odot$ are not impacted by the maximum duration cutoff. More details on this feature can be found at [26]. The minimum match [30] was set to 96.5% which ensures that no more than approximately 10% of astrophysical signals can be missed while limiting the total number of templates in the template bank. The detector’s noise PSD used to generate the template bank was obtained toward the end of O4a,

specifically, a week starting on December 29, 2023, at 00:00:00 UTC, to accurately reflect the sensitivity of the detectors in O4a. The waveform approximant used to generate templates was IMRPhenomD [74,75], which is a computationally efficient phenomenological waveform model that combines analytic post-Newtonian (PN) and effective one-body (EOB) methods for the inspiral portion of the waveforms and uses numerical relativity (NR) simulations for the merger and ringdown portion of the waveform of spin-aligned BBHs in the frequency domain. As a result, the O4 offline SSM template bank has 3452006 templates. Figure 1 shows the placement of the templates of the O4 offline SSM template bank.² We plot the bank in the space of component masses, m_1 - m_2 as well as in the parameter space of chirp mass, $\mathcal{M}$ [Eq. (2)] and effective spin, χ_{eff} [Eq. (1)] which are well-measured from GW signals.

$$\mathcal{M} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}. \quad (2)$$

IV. METHODS

As mentioned above, gravitational wave searches rely on template banks, which are collections of waveforms generated from specific physical parameters. Using geometric techniques, we reframe the bank construction problem to be the efficient covering of a subset of the intrinsic parameter space [31,68]. Specifically, an ideal bank may be viewed as a network of points in the physical parameter space distributed such that for any signal within the specified constraints, at least one template in the bank is sufficiently “near” the signal to produce a SNR above a target threshold [68]. Following standard geometric approaches [31], we define “nearness” using the intrinsic geometry of the parameter space, which will be detailed further in Sec. IV A. These techniques have been demonstrated to be effective for standard astrophysical parameters for compact binary systems [26]; however, in the lower-mass regions of parameter space necessary for a SSM search, problems arise in two areas of the treebank algorithm. First, numerical estimation of the MANIFOLD metric on which the geometric methods rely is more prone to instability. Second, there is insufficient template density near the boundary of the constraint submanifold leading to lower matches with extremal intrinsic parameters. We present two enhancements to the existing tree-bank algorithm that were necessary to produce a bank for the offline SSM search, namely, boundary enhancement and neighborhood metric estimation.

²The templates extend beyond the designed m_1 - m_2 or $\mathcal{M}$ - χ_{eff} parameter space as a result of MANIFOLD needing extra templates around the edges of the designed region to mitigate edge effects. Refer to [69] for details.

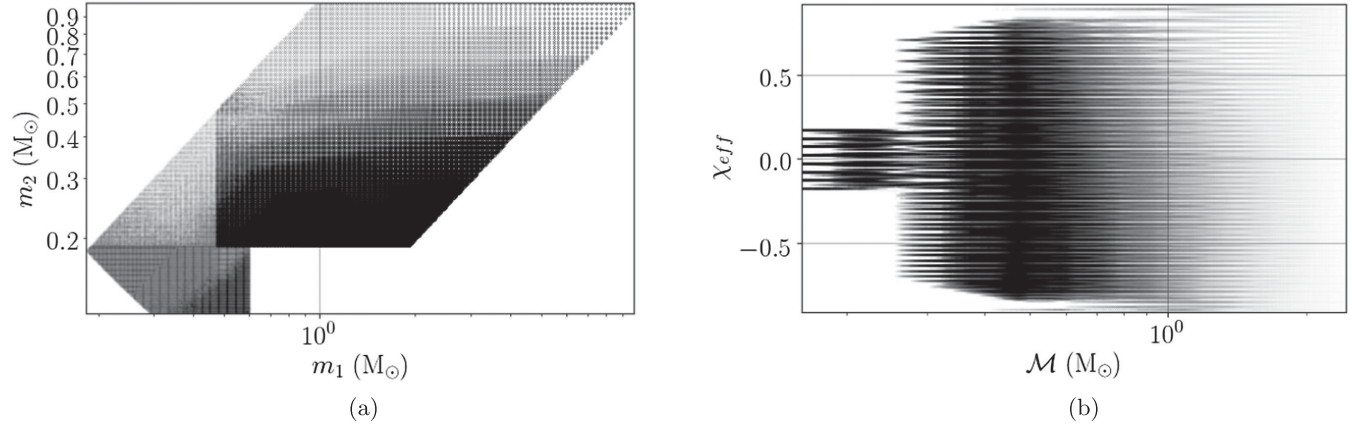


FIG. 1. Template placement of the O4 offline SSM template bank. (a) Template placement in the m_1 - m_2 space. (b) Template placement in the $\mathcal{M}$ - χ_{eff} space.

A. Geometric techniques and MANIFOLD

We begin our detailed discussion of methodology with a brief review of the geometric techniques used in MANIFOLD that are essential in the production of this bank [68]. The eponymous MANIFOLD $\mathfrak{M}$ is the set of possible waveforms for a given astrophysical event, such as CBC, which we parametrize using intrinsic-parameter coordinates such that a point p in $\mathfrak{M}$ is equivalent to a choice of waveform parameters $\vec{\lambda}$. Following standard waveform analysis, we define a measure of similarity of waveforms, called the *match*, which is essentially a correlation of the waveforms weighted inversely by detector noise and maximized over time-shifts and phases between the waveforms [68]. The match is computed as

$$\langle h|g \rangle \equiv \max_{t,\phi} \left| \int_{f_{\text{low}}}^{f_{\text{high}}} \frac{\tilde{h}(f) \tilde{g}^*(f)}{S_n(f)} df \right|, \quad (3)$$

where h and g are complex-valued time-domain waveforms containing both the sine and cosine phases, $\tilde{h}$ is the Fourier transform of h , $\tilde{g}^*$ is the complex-conjugate of $\tilde{g}$, and S_n is the PSD. We define the *mismatch*, denoted δ^2 to be

$$\delta^2(h, g) \equiv 1 - \langle h|g \rangle. \quad (4)$$

Noting that the mismatch functions similarly enough to a metric distance, we then define a Riemannian metric via the standard Fisher method [31]

$$g_{ij} = \partial_i \partial_j \delta^2, \quad (5)$$

which makes clear the notion of one signal being geometrically “near” to another, with the local identification of the distance d and the mismatch δ^2

$$d(h, g) \equiv \int_{\gamma} \sqrt{g_{\gamma(s)}[\gamma'(s), \gamma'(s)]} ds \approx \sqrt{\delta^2(h, g)}, \quad (6)$$

where γ is some smooth path connecting h and g in $\mathfrak{M}$ that minimizes the above integral. In regions $\mathcal{U} \subset \mathfrak{M}$ where the metric can be assumed to be constant, the geometric approach allows for fast computation of the mismatch as $\delta^2(h, g) = g_{ij} \Delta \tilde{\lambda}^i \Delta \tilde{\lambda}^j$ [68].

Further, the metric g also permits a local notion of *volume*. We define two types of volumes that are essential to the treebank algorithm: region volume and template volume. First, the volume of a for a small region, denoted $V_{\mathcal{U}}$, is computed following the standard Riemannian definition. For hyperrectangular regions $\mathcal{U}$ where the metric can be assumed to be constant, the volume may be efficiently computed using MANIFOLD as $V_{\mathcal{U}} = \sqrt{|g|} \prod_k \Delta \lambda^k$. Second, the *template volume* V_{δ^2} is defined in terms of a desired maximum mismatch between nearby templates [68], δ_{max}^2 . Assuming an n -dimensional parameter space, the template volume is given by $V_{\delta^2} = [2\sqrt{\delta_{\text{max}}^2/n}]^n$.

The treebank algorithm used by MANIFOLD produces a hyperrectangular partition of the parameter space in which each hyperrectangular region represents a template in the bank whose parameters are the center of the region [68]. The algorithm begins with one large hyperrectangle, iteratively splitting into smaller hyperrectangles following a binary tree technique, in which a region that is “too large” is split along its longest side according to the local metric. The notion of “too large” is defined where the volume of a hyperrectangular region R exceeds the desired template volume [68], or $V_R > V_{\delta^2}$. As a result, for each template h in the bank, there is a region $\mathcal{U}_h \subset \mathfrak{M}$ such that h is the only template contained in $\mathcal{U}_h$ and $V_{\mathcal{U}_h} < V_{\delta^2}$.³

³Note that the set $\{\mathcal{U}_h\}$ for all h in the bank constitutes a covering of the subset of $\mathfrak{M}$ specified by the bank constraints, and is the precise sense in which bank construction is a covering problem.

B. Metric estimation in SSM region

A principal assumption of the MANIFOLD method is the ability to estimate the local metric anywhere in the physically valid subset of the parameter space. Further, it is assumed that this metric has a Riemannian signature, or equivalently, all positive eigenvalues. These assumptions can break down in a region of the parameter space for several reasons, including decreased differentiability of the waveform function, degeneracy of one or more of the chosen coordinates, and others.

For the SSM region, exceedingly low masses caused both degeneracy of the χ_{eff} parameter as well as reduced smoothness of the waveform approximant IMRPhenomD, resulting in more frequent failure of the numerical estimation routines. These failures were detected by checking for negative eigenvalues. To remedy these failures, we generalized the notion of the local metric being estimated at a point to being estimated within a neighborhood of a point. We define the neighborhood of a point p in $\mathfrak{M}$ to be a local region $\mathcal{U}_p$ bounded by a maximum mismatch $\delta_{\mathcal{U}}^2$, such that for any other point q in $\mathcal{U}_p$, $\delta^2(p, q) < \delta_{\mathcal{U}}^2$. When it may be assumed the metric is locally constant, the neighborhood $\mathcal{U}_p$ takes the shape of a hyperellipsoid centered at p , for more on this construction see Appendix A.

The procedure for mismatch-neighborhood metric estimation at a point p , depicted in Fig. 2, amounts to the attempted metric estimation at each point q_i in a sequence of randomly selected points sampled uniformly over the neighborhood $\mathcal{U}_p$, terminating either on successful estimation of g or after a maximum number of attempts. Specifically, the sampling algorithm can be described by the following steps:

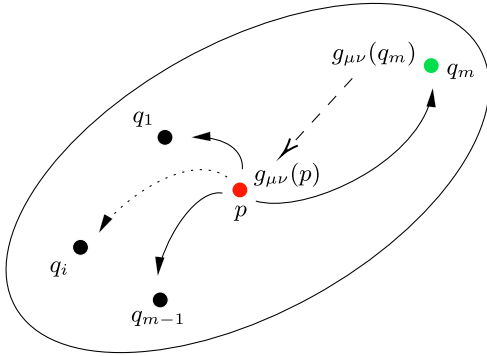


FIG. 2. Schematic representation of the bounded random sampling approach to solving metric estimation problems. The outer bounding ellipse represents a level surface of mismatch at p , using a previously evaluated metric from beyond the ellipse. Points q_k , $k \in \{1, \dots, m\}$ are randomly sampled within the ellipse, the first $m - 1$ iterations of which fail to resolve the error. Point q_m represents the first randomly sampled point to solve the error in metric evaluation. Last, the assumption is made $g_{\mu\nu}(p) \approx g_{\mu\nu}(q_m)$, which resolves the estimation error.

- (1) Metric evaluation fails at a point p , detected by the presence of negative eigenvalues in the metric g_{ij} .
- (2) A hyperellipsoid of maximum mismatch centered at p is determined using a reference metric.
- (3) A point q_i is randomly sampled from within the hyperellipsoid, so that q_i is “nearby” to p .
- (4) Metric estimation is attempted at q_i .
- (5) Repeat Steps 3–4 until the metric is successfully estimated at some point q_n , or until a maximum number of attempts is exceeded.
- (6) Finally, we use the metric $g(q_n)$ for the neighborhood by associating $g(p) \approx g(q_n)$.

Where the reference metric in Step 2 is produced by a general estimate at the beginning of the treebank algorithm and succeeded in each iteration of splitting by the metric of the parent hyperrectangle.

C. Boundary enhancement

Another principal assumption of the MANIFOLD technique is that templates within the original boundaries (the original hyperrectangle) are sufficient to recover signals in the same region. This assumption holds manifestly true in the bulk of the targeted search subset of the parameter space; however, it does not hold as well near the boundary of the given space. This agrees with the notion that a signal is best recovered when it is “surrounded” by templates [31], which emerges naturally from stochastic bank generation methods, as they often place templates outside the bounds of target signals [28]. The templates that are placed “outside” the target signal constraint space serve to improve the fitting factor to the most extreme targeted signals, e.g., those whose parameters are very near the maximum targeted values.

The MANIFOLD method used in [26] resulted in an insufficiently dense SSM template bank near the low-mass boundary of the constraint submanifold, both internally and externally to the constraint MANIFOLD, reducing the efficacy of the generated template bank in this parameter space. To improve the performance in these critical regions, we implemented a padding technique in which multiple banks with different constraint parameters can be combined along certain boundaries. The final bank was the union of two banks, the original bank as described above covering the targeted SSM parameter spaces, as well as a smaller “marginal” bank focused solely on the low-mass boundary. The process for finalizing a choice of padding-constraints, which were used to produce the marginal bank, was iterative, balancing the increase in number of templates against the performance of bank simulations along the low mass boundary. Specifically, the steps were (1) widen the constraints to cover more overlap at the boundary, (2) evaluate the signal recovery near the boundary, and (3) if the signal recovery was insufficient, repeat. In general, the more widely the constraints were padded, the more the

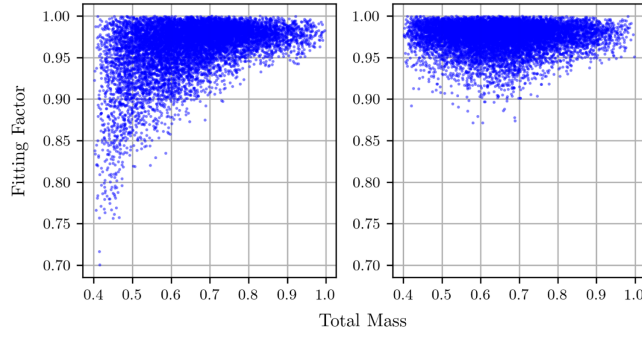


FIG. 3. Simulations of signal recovery for low mass events. The original bank (left) clearly has a lower recovery at the low mass boundary. The combined bank (right) has much greater performance at the low mass boundary.

improvement in the recovery of signals near the original boundary. We deemed an increase in the number of templates up to roughly 30% (on the order of 500,000 additional templates) to be an acceptable cost for improved coverage. We stopped widening the padding once the bank’s performance was sufficient—specifically, when a significant majority of simulated signals near the original boundary achieved match values above 95%. At that point, further increasing the bank yielded diminishing returns. Figure 3 shows the resulting impact of the boundary padding, in that signal recovery near the low mass boundary was dramatically improved.

V. RESULTS

Bank simulation is a method of computing the fitting factor (FF) between the template waveforms and simulated GW waveforms, to assess the effectualness of template banks against GW signals. The FF is given in [76,77] as

$$FF(\hat{u}_s, u) = \max_k \langle \hat{u}_k | \hat{u}_s \rangle \quad (7)$$

where $\langle \hat{u}_k | \hat{u}_s \rangle$ is defined in Eq. (3), u_s is the waveform of a simulated signal, $\hat{u}_i$ are normalized waveforms such that $\langle \hat{u}_i | \hat{u}_i \rangle = 1$, and k is the bank index such that $u = \{u_k\}_{k=1}^N$ and N is the number of templates in a bank.

To perform bank simulations with the presented banks, we use the SBank program, which simulates GW signals using LIGO Algorithm Library (LAL) [78] and computes the FF to find the template with the highest match for the signal. Previous work showed [26] that SBank bank simulation tests have been used to assess template banks generated with MANIFOLD, and that template banks generated with MANIFOLD are as effective as template banks generated with SBank.

In this paper, we present the bank simulation results of the O4 offline SSM bank using the SBank software [76,79–81]. We use the waveform approximant, the PSD, and the low-frequency cutoff described in Sec. III for

TABLE II. Parameter space constraints for simulated signals during bank testing for the three regions low-low, high-low, and high-high.

Parameter	Low-low	High-low	High-high
$m_1 (M_\odot)$	[0.2, 0.5]	[0.5, 10.0]	[0.5, 10.0]
$m_2 (M_\odot)$	[0.2, 0.5]	[0.2, 0.5]	[0.5, 1.0]
$s_{1,z}$	[-0.1, 0.1]	[-0.9, 0.9]	[-0.9, 0.9]
$s_{2,z}$	[-0.1, 0.1]	[-0.1, 0.1]	[-0.9, 0.9]
$f_{\text{low}}^{\text{cutoff}}$	75 Hz	55 Hz	45 Hz
$f_{\text{high}}^{\text{cutoff}}$	1024 Hz	1024 Hz	1024 Hz
Count	100,000	93,157	99,975

simulation studies presented here. It must be noted that MANIFOLD can apply a maximum duration when computing templates in its waveform computation, while SBank currently does not support this feature. Therefore, the bank simulation test below cannot reflect the SNR loss due to this duration cutoff in our templates. To estimate the bank performance while accounting for the maximum waveform duration cut applied during the template placement, we set a f_{low} cutoff higher than 45 Hz for different mass regions. See Table II.

We ran two bank simulation tests on the template bank as the following:

- (1) Assess the effectualness of the template bank by using simulated signals that span the template bank parameter space. (Sec. V A.)
- (2) Assess the effectualness of the template bank against electromagnetically bright events, i.e., BNSs and NSBHs. (Sec. V B.)

A. Simulation studies in the template bank space

The simulated signals used for efficacy tests presented in this section cover the same mass and spin parameter space as the template bank that is described in Sec. III. We choose to sample component masses from a log distribution since lower mass regions are more densely populated with templates. The template bank parameter space was divided into three regions for generating simulated signals, the details of which are summarized in Table II.

Figures 4–6 show the mismatches between the templates and the simulated signals in the three regions described above. 90% of the “low-low” simulated signals have a match of 97.77% or higher, 98.08% or higher for the “high-low” simulated signals, and 96.79% or higher for the “high-high” simulated signals. When the results of these three sets are combined, 90% of the simulated signals have a match of 97.42% or higher with templates in the bank. The low mismatches, i.e., higher FFs, of the “low-low” simulated signals are due to an over-density of templates in the extreme low-mass regions as a result of the padding procedure described in Sec. IV. In the high-high region, the relatively higher mismatches are a combined result of

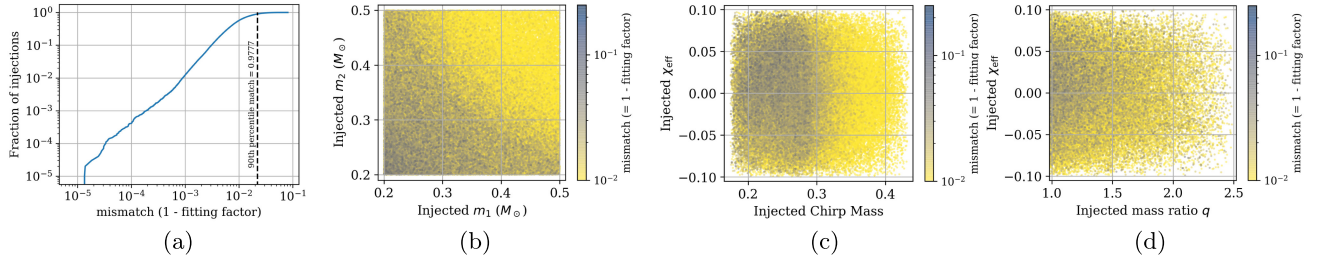


FIG. 4. Plots for simulated signals in the “low-low” region. For purposes of visually presenting the mismatches, mismatches smaller than 10^{-2} have been mapped to 10^{-2} . (a) shows that 90% of the simulated signals are recovered with a match $\geq 97.77\%$. (a) Cumulative histogram of mismatches. (b) Mismatches in the m_1 - m_2 plane. (c) Mismatches in the M - χ_{eff} plane. (d) Mismatches in the q - χ_{eff} plane.

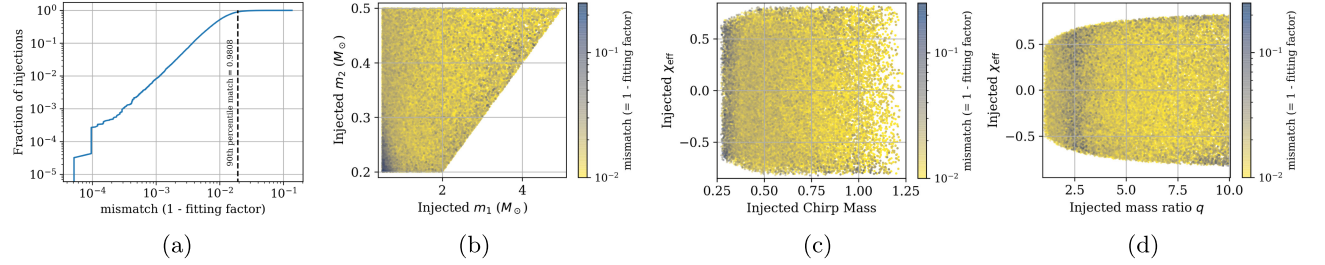


FIG. 5. Plots for the simulated signals in the “high-low” region. For purposes of visually presenting the mismatches, mismatches smaller than 10^{-2} have been mapped to 10^{-2} . (a) shows that 90% of the simulated signals are recovered with a match $\geq 98.08\%$. (a) Cumulative histogram of mismatches. (b) Mismatches in the m_1 - m_2 plane. (c) Mismatches in the M - χ_{eff} plane. (d) Mismatches in the q - χ_{eff} plane.

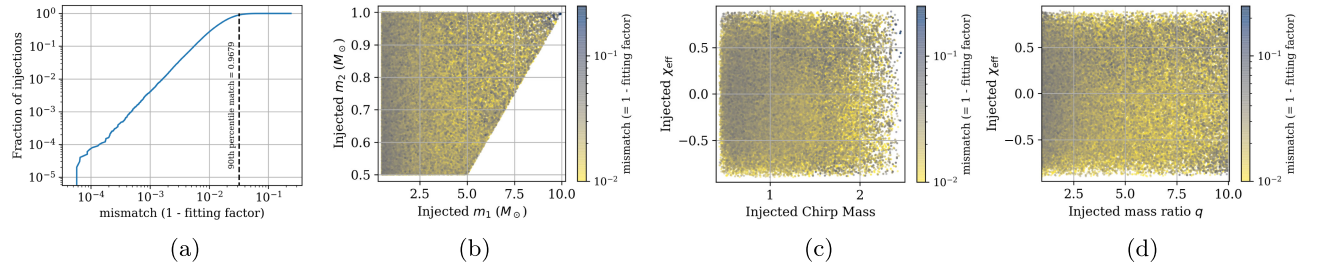


FIG. 6. Plots for the simulated signals in the “high-high” region. For purposes of visually presenting the mismatches, mismatches smaller than 10^{-2} have been mapped to 10^{-2} . (a) shows that 90% of the simulated signals are recovered with a match $\geq 96.79\%$. (a) Cumulative histogram of mismatches. (b) Mismatches in the m_1 - m_2 plane. (c) Mismatches in the M - χ_{eff} plane. (d) Mismatches in the q - χ_{eff} plane.

no padding and approaching a boundary of the template bank caused by the discontinuous spin constraints between components’ low and high masses. This particular boundary is naturally visible in the $\mathcal{M}$ - χ_{eff} plots, as in Fig. 1(b), and can be visualized as a vertical line separating the high and low spin regions. Despite the higher mismatches, the simulation studies show that the bulk of the high-high region is also densely populated with templates. The distribution of mismatches on combining the low-low, high-low, and high-high show that the templates populate

the search parameter space as designed and their density is sufficient to detect GW signals within the subset of parameter space of the SSM search.

In O4, the GstLAL pipeline runs the BNS/NSBH/BBH search as well as the SSM search, so multiple banks can detect the same GW signals that are at the edges of the template banks. We perform a bank simulation test to assess the effectualness of the search against such signals that lie at the edge of the O4 offline SSM template bank using the high-high set of simulated signals. For this study, we

TABLE III. Parameter space constraints for electromagnetically bright parameter regions.

Parameter	BNS low	BNS high	NSBH low	NSBH high
$m_1 (M_\odot)$	[1.0, 3.0]	[1.0, 3.0]	[3.0, 10.0]	[3.0, 10.0]
$m_2 (M_\odot)$	[1.0, 3.0]	[1.0, 3.0]	[1.0, 3.0]	[1.0, 3.0]
$s_{1,z}$	[-0.05, 0.05]	[-0.5, 0.5]	[-0.99, 0.99]	[-0.99, 0.99]
$s_{2,z}$	[-0.05, 0.05]	[-0.5, 0.5]	[-0.05, 0.05]	[-0.5, 0.5]
Count	100,000	100,000	100,000	100,000

combine the O4 BNS/NSBH/BBH template bank described in [26] and the O4 offline SSM.

Compared to the mismatches for the high-high signals with the SSM template bank, the combined bank gives a slightly higher match of 97.14% or higher for 90% of the simulated signals. The mismatches only decrease as the total mass increases as a result of having higher mass templates in the combined bank. This further illustrates that the SSM template bank sufficiently covers the parameter space of the SSM search.

B. Simulation studies with BNS and NSBH signals

In the m_1 - m_2 space, chirp mass forms an ambiguity contour leading to a degeneracy where binaries with different component masses can have the same chirp mass. Due to this, the chirp mass $\mathcal{M}$ of templates in the SSM

bank coincides with typical chirp masses of BNS and NSBH signals. Therefore, we perform efficacy tests in the BNS and NSBH parameter space in addition to the studies discussed in Sec. VA to assess the response of this bank toward electromagnetically bright events, which we define to be those in which at least one component is neutron star (NS). The parameters of the BNS and NSBH simulated signals are summarized in Table III. The motivations for limiting NS masses to $3.0M_\odot$ can be found in [26]. As done for the bank simulation tests presented in Sec. VA, we sample component masses from a log distribution.

We check the effectualness of the bank toward low-spinning and high-spinning BNS and NSBH populations. We use a lower frequency cutoff of 45 Hz for these bank simulation results. Figures 7–10 show the mismatches between the templates in the template bank and the BNS and NSBH simulated signals. With templates in the SSM bank, 90% of the BNS simulated signals with low-spin constraints have a match of 95.83% or higher, 95.85% or higher for BNS simulated signals with high-spin constraints, and 0% for low-spinning and high-spinning NSBH simulated signals. Figures 7–10 show that the BNS and NSBH simulated signals have low mismatches with the template bank in low $\mathcal{M}$ regions, but the mismatches spike above $\mathcal{M} \gtrsim 2.4M_\odot$. This is expected given that the highest $\mathcal{M}$ in the bank is $2.4M_\odot$. From these studies, we find that the O4 offline SSM template bank may recover GW signals

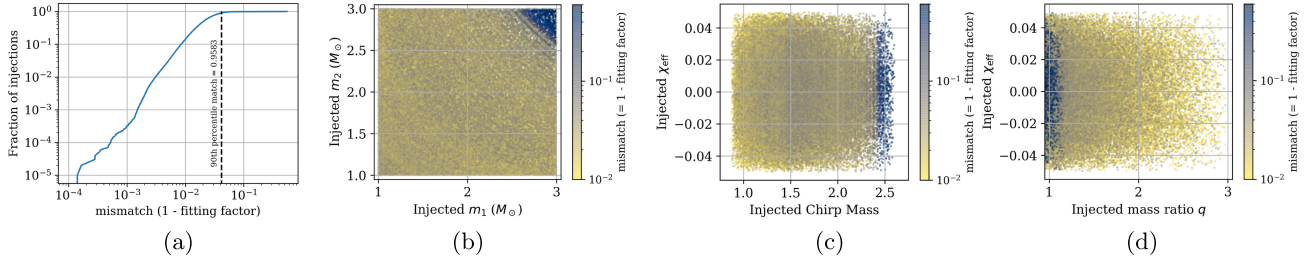


FIG. 7. Plots for BNS simulated signals with low-spins. For purposes of visually presenting the mismatches, mismatches smaller than 10^{-2} have been mapped to 10^{-2} . (a) shows that 90% of the simulated signals are recovered with a match $\geq 95.83\%$. (a) Cumulative histogram of mismatches. (b) Mismatches in the m_1 - m_2 plane. (c) Mismatches in the M - χ_{eff} plane. (d) Mismatches in the q - χ_{eff} plane.

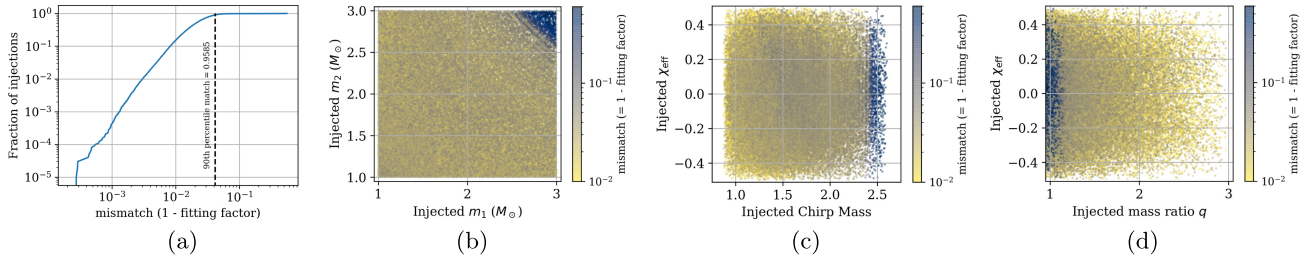


FIG. 8. Plots for BNS simulated signals with high-spins. For purposes of visually presenting the mismatches, mismatches smaller than 10^{-2} have been mapped to 10^{-2} . (a) shows that 90% of the simulated signals are recovered with a match $\geq 95.85\%$. (a) Cumulative histogram of mismatches. (b) Mismatches in the m_1 - m_2 plane. (c) Mismatches in the M - χ_{eff} plane. (d) Mismatches in the q - χ_{eff} plane.

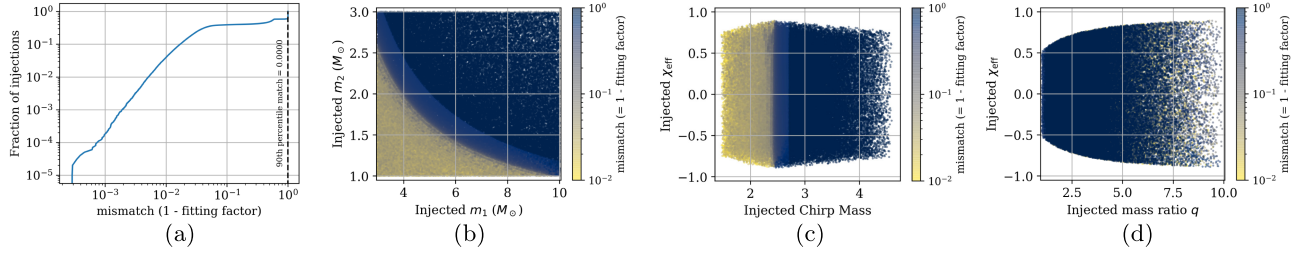


FIG. 9. Plots for NSBH simulated signals with low-spins. For purposes of visually presenting the mismatches, mismatches smaller than 10^{-2} have been mapped to 10^{-2} . (a) shows that 90% of the simulated signals are recovered with a match $\geq 0.00\%$. (a) Cumulative histogram of mismatches. (b) Mismatches in the m_1 - m_2 plane. (c) Mismatches in the M - χ_{eff} plane. (d) Mismatches in the q - χ_{eff} plane.

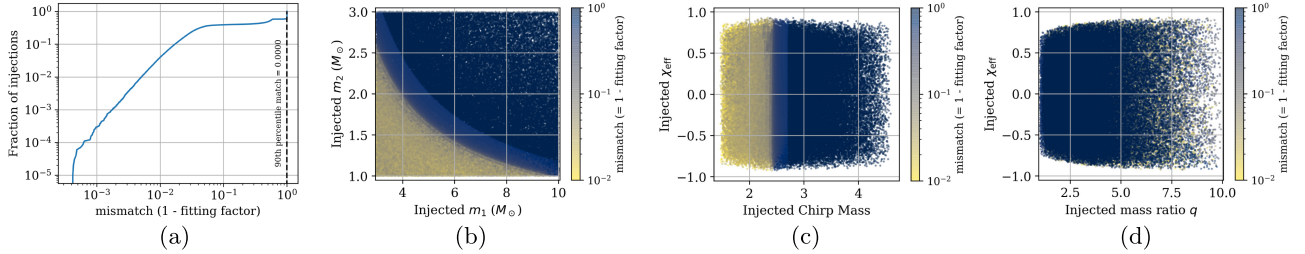


FIG. 10. Plots for NSBH simulated signals with high-spins. For purposes of visually presenting the mismatches, mismatches smaller than 10^{-2} have been mapped to 10^{-2} . (a) shows that 90% of the simulated signals are recovered with a match $\geq 0.00\%$. (a) Cumulative histogram of mismatches. (b) Mismatches in the m_1 - m_2 plane. (c) Mismatches in the M - χ_{eff} plane. (d) Mismatches in the q - χ_{eff} plane.

from electromagnetically bright events like BNSs and NSBHs that have $\mathcal{M} \lesssim 2.4M_\odot$.

VI. CONCLUSION

We have presented template banks intended for use in an offline and low-latency SSM searches as well as bank simulation results that show the banks to be sufficient for achieving their respective design objectives. These banks will be used by the GstLAL search pipeline in both low-latency (the bank in Appendix B) and archive (the “offline” bank in III) searches. Similar to the bank produced for the O4 low-latency GstLAL search [26], the banks presented in this paper make use of computationally efficient, geometric techniques from the MANIFOLD software. The offline bank for SSM searches presented in this paper spans the mass parameter space of 0.2 to $10M_\odot$ in the larger component and 0.2 to $1.0M_\odot$ in the smaller component, the spin parameter space of -0.9 to 0.9 in the larger component and -0.05 to 0.05 in the smaller component, and the mass ratio parameter space of 1 to 10 . The PSD used was from LIGO O4a and the low frequency cutoff was set to 45 Hz with a maximum waveform duration of 128 secs. The bank simulations have shown that the banks presented in this paper have sufficient efficacy for use in their respective searches. These banks will be used on data from the fourth observing run of Advanced LIGO, Virgo, and KAGRA.

Further, we have also shown required methodological improvements to the treebank algorithm in MANIFOLD necessary to produce a template bank for SSM searches, including enhanced numerical stability of metric estimation at exceedingly low masses and boundary padding for improving the recovery of the lowest mass signals in the search. The neighborhood metric estimation was effective in reducing metric estimation failures, as measure by the presence of negative eigenvalues in g . The boundary padding was effective in improving the lower matches along the low-mass boundary, as measured by the fitting factor during low-low simulations. These techniques have been added to MANIFOLD and will be available for future use should subsequent searches also want to cover lower-mass regions of the parameter space.

ACKNOWLEDGMENTS

This research has made use of data, software and/or web tools obtained from the Gravitational Wave Open Science Center [82], a service of LIGO Laboratory, the LIGO Scientific Collaboration (LSC) and the Virgo Collaboration. This material is based upon work supported by National Science Foundation (NSF)’s LIGO Laboratory which is a major facility fully funded by the NSF. We especially made heavy use of the LVK Algorithm Library [78,83]. LIGO was constructed by the California Institute of Technology and the Massachusetts Institute of

Technology with funding from the United States NSF and operates under cooperative agreements PHY-0757058 and PHY-0823459. In addition, the Science and Technology Facilities Council (STFC) of the United Kingdom, the Max-Planck-Society (MPS), and the State of Niedersachsen/Germany supported the construction of Advanced LIGO and construction and operation of the GEO600 detector. Additional support for Advanced LIGO was provided by the Australian Research Council. Virgo is funded, through the European Gravitational Observatory (EGO), by the French Centre National de Recherche Scientifique (CNRS), the Italian Istituto Nazionale di Fisica Nucleare (INFN) and the Dutch Nikhef, with contributions by institutions from Belgium, Germany, Greece, Hungary, Ireland, Japan, Monaco, Poland, Portugal, Spain. The authors are grateful for computational resources provided by the LIGO Lab cluster at the LIGO Laboratory and supported by Grants No. PHY-0757058 and No. PHY-0823459, the Pennsylvania State University's Institute for Computational and Data Sciences gravitational-wave cluster, and supported by Grants No. OAC-2103662, No. PHY-2308881, No. PHY-2011865, No. OAC-2201445, No. OAC-2018299, and No. PHY-2207728. C.H. acknowledges generous support from the Eberly College of Science, the Department of Physics, the Institute for Gravitation and the Cosmos, the Institute for Computational and Data Sciences, and the Freed Early Career Professorship. We thank Ines Bentara for their helpful comments and suggestions.

DATA AVAILABILITY

The data that support the findings of this article are openly available [84–86].

APPENDIX A: METRIC HYPERELLIPSES

1. Necessary assumptions for hyperellipsoids

In general, a level surface of a Riemannian metric cannot be assumed to be an ellipsoid, as variations in curvature over the MANIFOLD could result in an arbitrarily warped, smooth, closed surface. However, within a local ball of radius r such that $r^2 \ll \frac{1}{R}$ where R is the Ricci curvature, we assume the metric to be locally constant. In this context, the metric is constant, symmetric, and positive definite, which is sufficient to describe a hyperellipsoid. The canonical form of the equation for a hyperellipsoid is given as

$$\frac{x_1^2}{a_1^2} + \frac{x_2^2}{a_2^2} + \cdots + \frac{x_n^2}{a_n^2} = 1 \quad (\text{A1})$$

Similarly, a hyperellipsoid $\mathcal{E}$ can be constructed by deforming an n -sphere using linear transformations along each principle axis. We generalize the above canonical description to more broadly define a hyperellipsoid as

$$\mathcal{E} \equiv \{p | A_{\mu\nu} x^\mu(p) x^\nu(p) = C\} \quad (\text{A2})$$

where A is a constant matrix that is positive semidefinite and $C \in \mathbb{R}$ is a constant, often chosen as $C = 1$ for convenience. In cases related to metric distance, this constant is chosen to be the defining distance squared of the surface. Note that for bank creation, the mismatch is already a squared-distance.

2. Eigenvalue decomposition and principal axes

The eigenvalue decomposition finds solutions to the eigenvalue equation, defined below, of the given matrix A .

$$A_{\mu\nu} v^\nu = \lambda v^\mu \quad (\text{A3})$$

This is achieved by diagonalizing the matrix A , which results in a diagonal matrix of eigenvalues Λ . Note that $\Lambda_{ij} = \delta_{ij} \lambda_j$, where λ_i is the eigenvalue corresponding to the i th solution of the eigenvalue equation, v_i^μ . In the basis of eigenvectors, the generalized ellipsoid definition in Eq. (A2) reduces to the canonical form in Eq. (A1).

$$\begin{aligned} 1 &= \Lambda_{\mu\nu} x^\mu x^\nu = \delta_{\mu\nu} x^\mu x^\nu \\ &= \lambda_1 (x^1)^2 + \cdots + \lambda_n (x^n)^2 \\ &= \frac{(x^1)^2}{(a^1)^2} + \cdots + \frac{(x^n)^2}{(a^n)^2} \end{aligned} \quad (\text{A4})$$

where $a^i \equiv \sqrt{1/\lambda^i}$ is a vector of scale factors with index i referring to the i th eigenvector v_i^μ . Using these scale factors, it is possible to recover the principal axis vectors p_i^μ of the ellipsoid, where $p_i^\mu \equiv a_i v_i^\mu$. The principal axis vectors have the virtue of being orthogonal in the ambient space, which can be more natural or intuitive when dealing with ellipsoid geometry. They provide a natural basis for reconstructing ellipsoid points efficiently in code by requiring fewer matrix operations.

APPENDIX B: TEMPLATE BANK FOR LOW-LATENCY SSM SEARCH

In previous observing runs [42–45], the GstLAL pipeline searched for binaries with at least one SSM component after archival data was available. In O4, GstLAL performs SSM searches in near real-time, i.e., low-latency, motivated by the discovery of a previously unobserved class of compact objects along with the possibility of detecting electromagnetically bright mergers that lie outside the search parameter space of typical searches for compact binary mergers. This search is the first low-latency SSM search performed by the GstLAL pipeline. In this section, the template bank used for this low-latency search is discussed.

For the low-latency search, we continue to target systems with at least one SSM compact object. Since points in the

TABLE IV. Parameter space of the template bank used in O4a low-latency SSM search by GstLAL.

Parameter	GstLAL
Primary mass, m_1	$\in [0.5, 10M_\odot]$
Secondary Mass, m_2	$\in [0.5, 1M_\odot]$
Mass ratio, $q = m_1/m_2$	$\in [1, 10]$
Dimensionless spin, $s_{i,z}$	$ s_{i,z} < 0.3$
Lower frequency cutoff	45 Hz
Higher frequency cutoff	1024 Hz
Waveform approximant	IMRPhenomD
Minimum match	0.97
Maximum waveform duration	128 secs
Total number of templates	1,069,651

mass parameter space become more correlated at lower masses, a higher number of templates is needed to optimally cover the target region as the minimum component masses are lowered. To mitigate this, the smallest component masses are restricted to $0.5M_\odot$ for the low-latency search, unlike in the archival searches, i.e., offline searches, where component masses go down to $0.2M_\odot$. Since the sensitivity of the search increases with the chirp mass of BBHs [45], the choice of a higher minimum component mass restricts the search to higher VT regions of the parameter space.

Like all archival SSM searches, only spin-aligned templates are included. Therefore, each template only has nonzero z-component spins for each object which contribute to the effective spin parameter, χ_{eff} . Additionally, MANIFOLD [73] assigns equal z-component spins to both components such that $\chi_{\text{eff}} = s_{i,z}$. The effective spin parameter, $\chi_{\text{eff}} (= s_{i,z})$, is further restricted to ± 0.3 based on [73] where it is shown that short duration non-stationarities in strain data have $\chi_{\text{eff}} < -0.3$.

Table IV lists the parameter space and configuration that was used in the construction of this bank, which is visualized by Fig. 11.

1. Effectualness of template bank

Bank simulation studies were performed to determine if the constructed bank efficiently covers the target search space. Since the mass boundaries extend above a solar mass for the primary, this search is sensitive to BNS and NSBH mergers in addition to mergers of SSM black holes. Therefore, we perform bank simulations for the following distinct populations

- (i) SSM-BBH: primary target of this search—BBH mergers with $m_1 \in [0.5, 10M_\odot]$, $m_2 \in [0.5, 1M_\odot]$, and $|\chi_{\text{eff}}| < 0.3$.
- (ii) BNS-LOW: BNS mergers with $m_{\text{NS}} \in [1, 3M_\odot]$ and $|\chi_{\text{eff}}| < 0.05$.
- (iii) BNS-HIGH: BNS mergers with $m_{\text{NS}} \in [1, 3M_\odot]$ and $|\chi_{\text{eff,NS}}| < 0.5$.

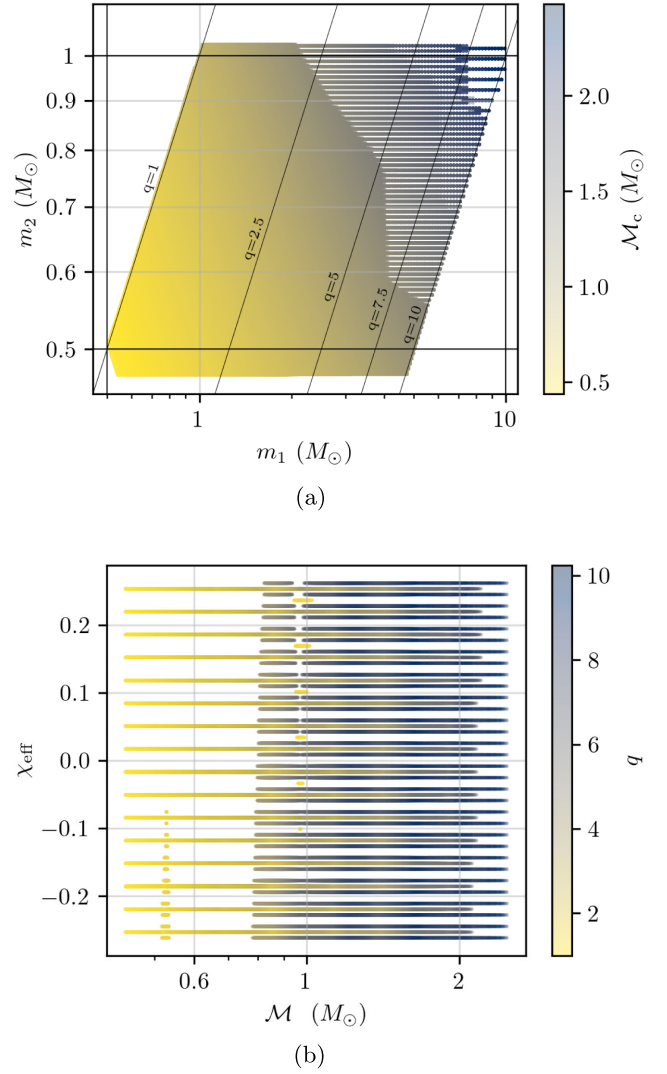


FIG. 11. Template placement in the O4 low-latency SSM bank. The color maps in (a) and (b) show trends in chirp mass, $\mathcal{M}$ and mass-ratio, q respectively. (a) Template placement in the m_1 - m_2 space. (b) Template placement in the $\mathcal{M}$ - χ_{eff} space.

- (iv) NSBH-LOW: NSBH mergers with $m_{\text{NS}} \in [1, 3M_\odot]$, $m_{\text{BH}} \in [3, 10M_\odot]$, $|\chi_{\text{eff,NS}}| < 0.05$, and $|\chi_{\text{eff,BH}}| < 0.99$.
- (v) NSBH-HIGH: NSBH mergers with $m_{\text{NS}} \in [1, 3M_\odot]$, $m_{\text{BH}} \in [3, 10M_\odot]$, $|\chi_{\text{eff,NS}}| < 0.5$, and $|\chi_{\text{eff,BH}}| < 0.99$.

Since the bank is constructed with a required minimal match of 0.97, the expected highest mismatch for most SSM-BBH signals is ≤ 0.03 . As shown in Fig. 12(a), we find that 90% of the injections are recovered with a match of 0.99. The bank is less effectual at the boundaries of the bank due to difficulties with template placement at the edges with MANIFOLD. This is mitigated by extending the boundaries of the bank during construction beyond the desired target region.

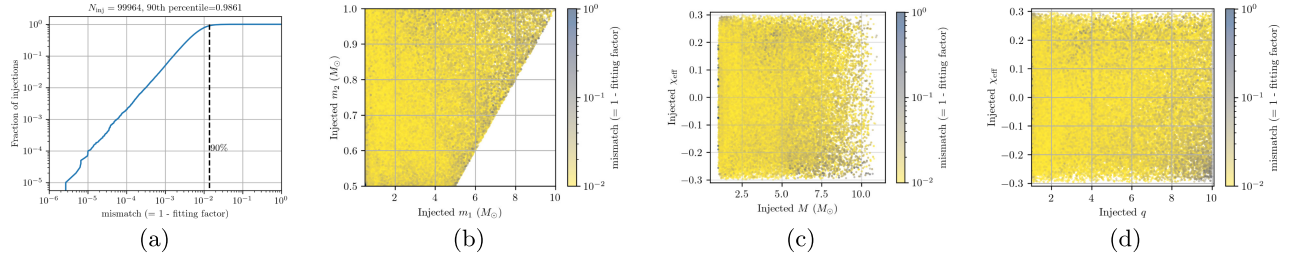


FIG. 12. Plots for the SSM-BBH injections. (a) shows that 90% of the injections are recovered with a match ≥ 0.99 . (a) Cumulative histogram of mismatches. (b) Mismatches in the m_1 - m_2 plane. (c) Mismatches in the M - χ_{eff} plane. (d) Mismatches in the q - χ_{eff} plane.

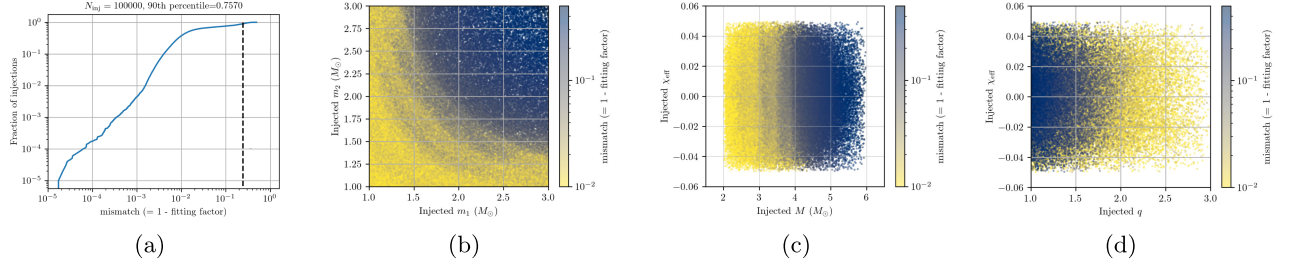


FIG. 13. Plots for the BNS-LOW injections. (a) shows that 90% of the injections are recovered with a match ≥ 0.76 . (a) Cumulative histogram of mismatches. (b) Mismatches in the m_1 - m_2 plane. (c) Mismatches in the M - χ_{eff} plane. (d) Mismatches in the q - χ_{eff} plane.

BNS-LOW and NSBH-LOW represent astrophysically expected binary systems where the NSs are extremely low-spinning when they occur in a binary. In the BNS-LOW regime, this bank effectively recovers 90% of the injections with a match ≥ 0.76 . Figure 13 shows that more asymmetric BNS with low total mass, M are detectable with this template bank irrespective of χ_{eff} . For typical NSBH systems, NSBH-LOW, the bank performs suboptimally with most injections lying completely outside the target space of the search Fig. 15. Only 10% of the injections have mismatches ≤ 0.03 , and lie mostly in $-0.75 \leq \chi_{\text{eff}} \leq 0.25$ and $4M_\odot \leq M \leq 7M_\odot$ with a cutoff at $M \approx 2.5M_\odot$ which is the maximum chirp mass in the template bank. Here, M is the total mass, $M = m_1 + m_2$.

The low spin limit for NSs in binaries is primarily informed by observations of the fastest-spinning pulsars

in BNS systems which are expected to merge within Hubble time—PSR J0737-3039A [87] and PSR J1946 + 2052 [88]. These systems are expected to evolve to $\chi < 0.04$ and $\chi < 0.05$ at the time of the merger. However, it is within the realm of possibility that millisecond pulsars form binaries but these have not yet been observed. In [89], it is also shown that if the NS forms before the BH in an NSBH, the resultant spins could be much higher than 0.05. Considering that the BNS/NSBH/BBH searches do not target high-spinning NSs in binaries and the O4 template bank used by GstLAL is ineffectual in recovering such high spin BNSs [26], the SSM search provides a unique opportunity to detect previously unexplored NSs with high spins. Therefore, we perform bank simulations with BNS-HIGH and NSBH-HIGH signals shown in Figs. 14 and 16. We find that 90% of BNS-HIGH injections are recovered with a

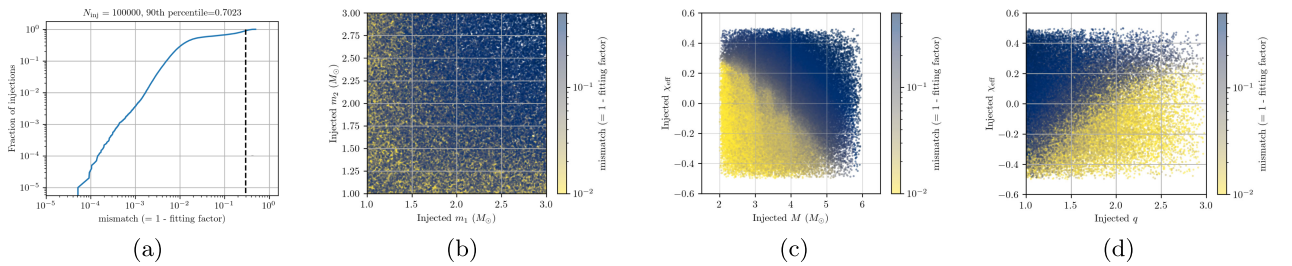


FIG. 14. Plots for the BNS-HIGH injections. (a) shows that 90% of the injections are recovered with a match ≥ 0.70 . (a) Cumulative histogram of mismatches. (b) Mismatches in the m_1 - m_2 plane. (c) Mismatches in the M - χ_{eff} plane. (d) Mismatches in the q - χ_{eff} plane.

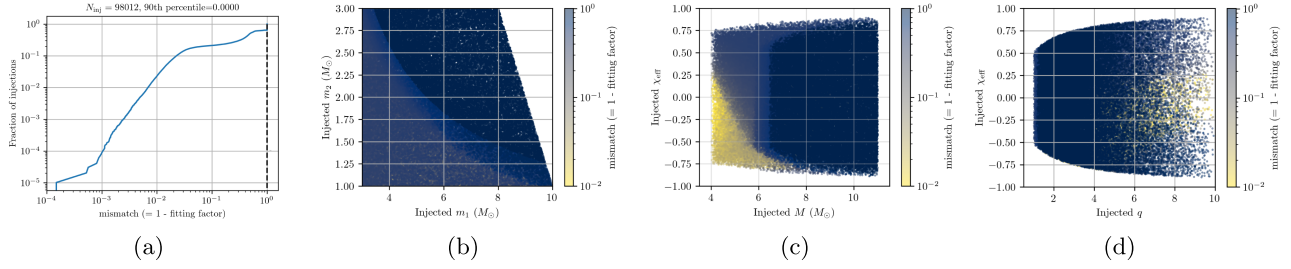


FIG. 15. Plots for the NSBH-LOW injections. (a) shows that only 10% injections are recovered with a mismatch ≤ 0.03 . (a) Cumulative histogram of mismatches. (b) Mismatches in the m_1 - m_2 plane. (c) Mismatches in the M - χ_{eff} plane. (d) Mismatches in the q - χ_{eff} plane.

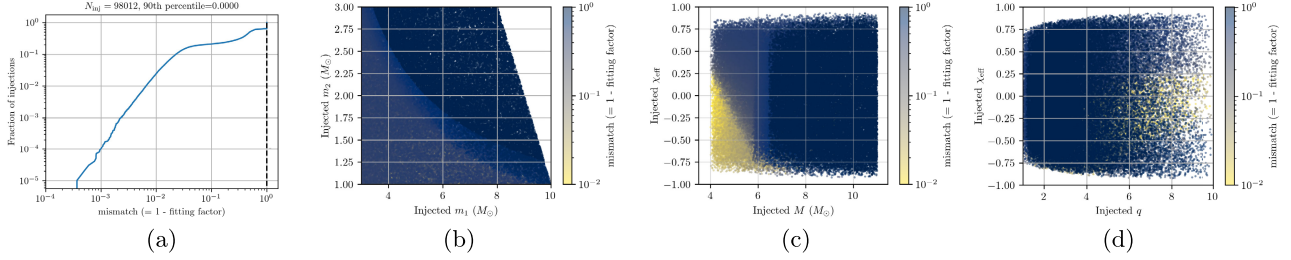


FIG. 16. Plots for the NSBH-HIGH injections. (a) shows that only 10% injections are recovered with a mismatch ≤ 0.03 . (a) Cumulative histogram of mismatches. (b) Mismatches in the m_1 - m_2 plane. (c) Mismatches in the M - χ_{eff} plane. (d) Mismatches in the q - χ_{eff} plane.

match of 0.7. On the other hand, the bank is not effectual in the NSBH-HIGH regime which is evident in Fig. 16(a). Similar to NSBH-LOW, only 10% of the injections have mismatches ≤ 0.03 , which mostly have $-0.8 \leq \chi_{\text{eff}} \leq 0.25$ and $4M_{\odot} \leq M \leq 7M_{\odot}$ with a cutoff at $\mathcal{M} \approx 2.5M_{\odot}$ which is the maximum chirp mass in the template bank.

2. Population model

The population model was pregenerated using MANIFOLD. An uninformative prior is used to generate the population model ensuring that all templates in the bank carry uniform weight due to a lack of prior detections. We disregard expected rates from models that predict SSM compact objects to remove any bias from the search. The population model used in this search is shown in Fig. 17. This model is only used in likelihood estimation here, since we do not yet assign source classification probabilities to events from this search.

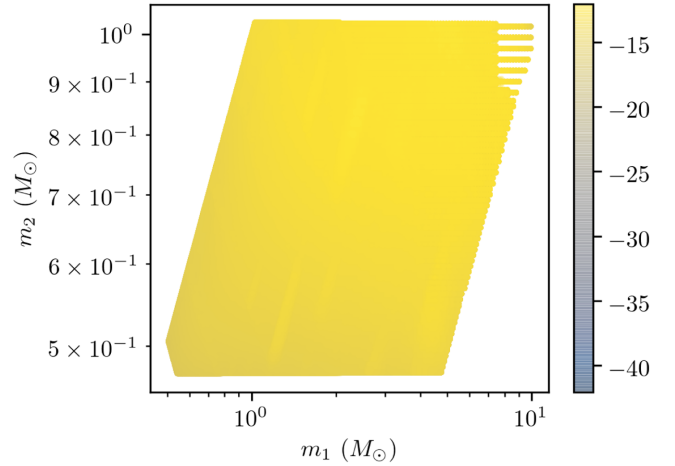


FIG. 17. Template weights corresponding to the uniform population distribution, as a function of component masses. The color bars represent $\log P(t_k, \rho_k | H_{\text{SSM}})$ for $\rho_k = 10$.

- [1] B. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), *Phys. Rev. Lett.* **116**, 061102 (2016).
 [2] B. P. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), *Phys. Rev. Lett.* **116**, 131103 (2016).

- [3] B. P. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), *Phys. Rev. X* **9**, 031040 (2019).
 [4] G. M. Harry (LIGO Scientific Collaboration), *Classical Quantum Gravity* **27**, 084006 (2010).

- [5] LIGO Scientific Collaboration, *Classical Quantum Gravity* **32**, 074001 (2015).
- [6] M. Tse *et al.*, *Phys. Rev. Lett.* **123**, 231107 (2019).
- [7] A. Buikema *et al.*, *Phys. Rev. D* **102**, 062003 (2020).
- [8] E. Capote *et al.*, *Phys. Rev. D* **111**, 062002 (2025).
- [9] F. Acernese *et al.* (Virgo Collaboration), *Classical Quantum Gravity* **32**, 024001 (2015).
- [10] F. Acernese *et al.* (Virgo Collaboration), *Phys. Rev. Lett.* **123**, 231108 (2019).
- [11] B. P. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), *Phys. Rev. Lett.* **119**, 161101 (2017).
- [12] B. P. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), *Astrophys. J. Lett.* **892**, L3 (2020).
- [13] B. P. Abbott *et al.* (LIGO Scientific, Virgo, and KAGRA Collaborations), *Astrophys. J. Lett.* **915**, L5 (2021).
- [14] B. P. Abbott *et al.* (LIGO Scientific, Virgo, and KAGRA Collaborations), *Phys. Rev. X* **13**, 011048 (2023).
- [15] B. P. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), *Phys. Rev. Lett.* **121**, 161101 (2018).
- [16] B. P. Abbott *et al.* (LIGO Scientific, Virgo, and KAGRA Collaborations), *Living Rev. Relativity* **23**, 3 (2020).
- [17] T. Akutsu *et al.* (KAGRA Collaboration), *Prog. Theor. Exp. Phys.* **2021**, 05A103 (2021).
- [18] T. Akutsu *et al.*, *Prog. Theor. Exp. Phys.* **2021**, 05A101 (2020).
- [19] R. Abbott *et al.* (LIGO Scientific, Virgo, and KAGRA Collaborations), *Phys. Rev. X* **13**, 041039 (2023).
- [20] Gstreamer plugins for the LSC algorithm library, <https://www.lsc-group.phys.uwm.edu/daswg/projects/gstlal.html> (2016).
- [21] S. Sachdev *et al.*, [arXiv:1901.08580](https://arxiv.org/abs/1901.08580).
- [22] C. Hanna *et al.*, *Phys. Rev. D* **101**, 022003 (2020).
- [23] K. Cannon *et al.*, *SoftwareX* **14**, 100680 (2021).
- [24] B. Ewing *et al.*, *Phys. Rev. D* **109**, 042008 (2024).
- [25] L. Tsukada *et al.*, *Phys. Rev. D* **108**, 043004 (2023).
- [26] S. Sakon *et al.*, *Phys. Rev. D* **109**, 044066 (2024).
- [27] A. Ray *et al.*, [arXiv:2306.07190](https://arxiv.org/abs/2306.07190).
- [28] B. S. Sathyaprakash and S. V. Dhurandhar, *Phys. Rev. D* **44**, 3819 (1991).
- [29] S. V. Dhurandhar and B. S. Sathyaprakash, *Phys. Rev. D* **49**, 1707 (1994).
- [30] B. J. Owen, *Phys. Rev. D* **53**, 6749 (1996).
- [31] B. J. Owen and B. S. Sathyaprakash, *Phys. Rev. D* **60**, 022002 (1999).
- [32] S. A. Usman *et al.*, *Classical Quantum Gravity* **33**, 215004 (2016).
- [33] A. H. Nitz, T. Dent, T. D. Canton, S. Fairhurst, and D. A. Brown, *Astrophys. J.* **849**, 118 (2017).
- [34] A. H. Nitz, T. Dal Canton, D. Davis, and S. Reyes, *Phys. Rev. D* **98**, 024050 (2018).
- [35] T. D. Canton, A. H. Nitz, B. Gadre, G. S. C. Davies, V. Villa-Ortega, T. Dent, I. Harry, and L. Xiao, *Astrophys. J.* **923**, 254 (2021).
- [36] T. Adams, D. Buskulic, V. Germain, G. M. Guidi, F. Marion, M. Montani, B. Mours, F. Piergiovanni, and G. Wang, *Classical Quantum Gravity* **33**, 175012 (2016).
- [37] F. Aubin, F. Brighenti, R. Chierici, D. Estevez, G. Greco, G. M. Guidi, V. Juste, F. Marion, B. Mours, E. Nitoglia, O. Sauter, and V. Sordini, *Classical Quantum Gravity* **38**, 095004 (2021).
- [38] N. Andres, M. Assiduo, F. Aubin, R. Chierici, D. Estevez, F. Faedi, G. M. Guidi, V. Juste, F. Marion, B. Mours, E. Nitoglia, and V. Sordini, *Classical Quantum Gravity* **39**, 055002 (2022).
- [39] C. All  n  , F. Aubin, I. Bentara, D. Buskulic, G. M. Guidi, V. Juste, M. Lethuillier, F. Marion, L. Mobilia, B. Mours, A. Ouzriat, T. Sainrat, and V. Sordini, *Classical Quantum Gravity* **42**, 105009 (2025).
- [40] B. Abbott *et al.* (LIGO Scientific Collaboration), *Phys. Rev. D* **72**, 082002 (2005).
- [41] B. Abbott *et al.* (LIGO Scientific Collaboration), *Phys. Rev. D* **77**, 062002 (2008).
- [42] B. P. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), *Phys. Rev. Lett.* **121**, 231103 (2018).
- [43] B. P. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), *Phys. Rev. Lett.* **123**, 161102 (2019).
- [44] B. P. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), *Phys. Rev. Lett.* **129**, 061104 (2022).
- [45] LIGO Scientific, Virgo, and KAGRA Collaborations, *Mon. Not. R. Astron. Soc.* **524**, 5984 (2023); **526**, 6234(E) (2023).
- [46] A. H. Nitz and Y.-F. Wang, *Astrophys. J.* **915**, 54 (2021).
- [47] A. H. Nitz and Y.-F. Wang, *Phys. Rev. Lett.* **126**, 021103 (2021).
- [48] A. H. Nitz and Y.-F. Wang, *Phys. Rev. Lett.* **127**, 151101 (2021).
- [49] A. H. Nitz and Y.-F. Wang, *Phys. Rev. D* **106**, 023024 (2022).
- [50] K. S. Phukon, G. Baltus, S. Caudill, S. Clesse, A. Depasse, M. Fays, H. Fong, S. J. Kapadia, R. Magee, and A. J. Tanasijczuk, [arXiv:2105.11449](https://arxiv.org/abs/2105.11449).
- [51] B. Carr, K. Kohri, Y. Sendouda, and J. Yokoyama, *Rep. Prog. Phys.* **84**, 116902 (2021).
- [52] B. Carr and F. Kuhnel, *SciPost Phys. Lect. Notes* **48**, 1 (2022).
- [53] B. P. Abbott *et al.*, *Phys. Rev. Lett.* **116**, 061102 (2016).
- [54] S. Clesse and J. Garcia-Bellido, *Phys. Dark Universe* **15**, 142 (2016).
- [55] M. Sasaki, T. Suyama, T. Tanaka, and S. Yokoyama, *Phys. Rev. Lett.* **117**, 061101 (2018).
- [56] S. Shandera, D. Jeong, and H. Gebhardt, *Phys. Rev. Lett.* **120**, 241102 (2018).
- [57] M. Ryan, S. Shandera, J. Gurian, and J. D., *Astrophys. J.* **934**, 122 (2022).
- [58] D. Singh, M. Ryan, R. Magee, T. Akhter, S. Shandera, D. Jeong, and C. Hanna, *Phys. Rev. D* **104**, 044015 (2021).
- [59] F. S. E. W. Mielke, *Nucl. Phys.* **B564**, 185 (2000).
- [60] C. Kouvaris and P. Tinyakov, *Phys. Rev. D* **82**, 063531 (2010).
- [61] S. D. McDermott, H.-B. Yu, and K. M. Zurek, *Phys. Rev. D* **85**, 023519 (2012).
- [62] J. Bramante and T. Linden, *Phys. Rev. Lett.* **113**, 191301 (2014).
- [63] D. Singh, A. Gupta, E. Berti, S. Reddy, and B. S. Sathyaprakash, *Phys. Rev. D* **107**, 083037 (2023).
- [64] S. Bhattacharya, B. Dasgupta, B. Laha, and A. Ray, *Phys. Rev. Lett.* **131**, 091401 (2023).
- [65] B. Dasgupta, R. Laha, and A. Ray, *Phys. Rev. Lett.* **126**, 141105 (2021).
- [66] R. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), *Phys. Rev. X* **11**, 021053 (2021).

- [67] R. Abbott *et al.* (LIGO Scientific and Virgo Collaborations), *Phys. Rev. X* **11**, 021053 (2021).
- [68] C. Hanna *et al.*, *Phys. Rev. D* **108**, 042003 (2023).
- [69] C. Hanna, MANIFOLD, <https://git.ligo.org/chad-hanna/manifold> (2024).
- [70] H. K. Y. Fong, From simulations to signals: Analyzing gravitational waves from compact binary coalescences, Ph.D. thesis, University of Toronto (Canada), 2018.
- [71] S. Schmidt *et al.*, *Phys. Rev. D* **110**, 023038 (2024).
- [72] S. Schmidt *et al.*, [arXiv:2306.07190](https://arxiv.org/abs/2306.07190).
- [73] C. Hanna *et al.*, *Phys. Rev. D* **106**, 084033 (2022).
- [74] S. Khan, S. Husa, M. Hannam, F. Ohme, M. Pürrer, X. J. Forteza, and A. Bohé, *Phys. Rev. D* **93**, 044007 (2016).
- [75] S. Husa, S. Khan, M. Hannam, M. Pürrer, F. Ohme, X. J. Forteza, and A. Bohé, *Phys. Rev. D* **93**, 044006 (2016).
- [76] S. Privitera, S. R. P. Mohapatra, P. Ajith, K. Cannon, N. Fotopoulos, M. A. Frei, C. Hanna, A. J. Weinstein, and J. T. Whelan, *Phys. Rev. D* **89**, 024003 (2014).
- [77] T. A. Apostolatos, *Phys. Rev. D* **52**, 605 (1995).
- [78] LIGO Scientific Collaboration, LIGO Algorithm Library—LALSuite, free software (GPL) (2018).
- [79] P. Ajith, N. Fotopoulos, S. Privitera, A. Neunzert, N. Mazumder, and A. Weinstein, *Phys. Rev. D* **89**, 084041 (2014).
- [80] C. Capano, I. Harry, S. Privitera, and A. Buonanno, *Phys. Rev. D* **93**, 124007 (2016).
- [81] I. W. Harry, B. Allen, and B. S. Sathyaprakash, *Phys. Rev. D* **80**, 104014 (2009).
- [82] <https://www.gw-openscience.org/>
- [83] LSC Algorithm Library software packages LAL, LALWrapper, and LALApps.
- [84] <https://www.gw-openscience.org>
- [85] <https://git.ligo.org/chad-hanna/manifold>
- [86] <https://dcc.ligo.org/T2400099-v3>
- [87] A. G. Lyne, *Chin. J. Astron. Astrophys.* **6**, 162 (2006).
- [88] K. Stovall *et al.*, *Astrophys. J. Lett.* **854**, L22 (2018).
- [89] D. Chattopadhyay, S. Stevenson, J. R. Hurley, M. Bailes, and F. Broekgaarden, *Mon. Not. R. Astron. Soc.* **504**, 3682 (2021).